



API & SMTP Verification Report

DVein HRM Web · backend, staff console, public registration, and Gmail SMTP

**SOURCE
READY**

All locally executable checks pass. Live Gmail delivery, deployed Cloud Run state, and Firebase emulator integration remain externally verifiable items.

41	54	97	20
API paths	API operations	backend checks passed	frontend tests passed

Executive conclusion

The reorganized project is internally consistent and buildable. The FastAPI process boots, its health endpoint and generated OpenAPI document respond over real local HTTP, both React applications lint and build, and their compiled artifacts serve successfully. The SMTP implementation is configured for dveininnovation@gmail.com, validates Google-specific security rules, uses certificate-verified TLS, and passes local delivery and MIME tests.

A live connection to smtp.gmail.com:587 was attempted twice (readiness and self-delivery commands). This workspace could not resolve the SMTP hostname and returned socket.gaierror before TLS or login. Therefore the supplied App Password was neither accepted nor rejected, and no test email was sent. Live delivery must not be represented as passed until the included command succeeds from a network that permits outbound SMTP.

Assessment at a glance

Area	Result	Meaning
Source and builds	PASS	Backend and both frontends passed all available local checks.
SMTP implementation	PASS	Configuration, TLS branches, authentication flow, MIME output, and catcher delivery are tested.
Live Gmail	BLOCKED	Workspace DNS/network policy prevented reaching smtp.gmail.com; credentials remain unverified.
Firebase integration	NOT RUN	213 tests require Firestore/Storage emulator binaries unavailable in this environment.
Production service	NOT CHANGED	No Cloud Run credentials/tooling were available; safe deployment commands are documented.

Changes completed

- Configured the ignored local environment for smtp.gmail.com, STARTTLS, port 587, and the complete sender/reply-to identity dveininnovation@gmail.com.
- Updated the production environment template with non-secret Gmail values; SMTP_PASSWORD remains absent and is intended for Google Secret Manager / Cloud Run secret mounting.
- Added Gmail configuration validation: username/password must be paired, STARTTLS and SSL must use their correct ports, insecure transport is rejected, and display spaces are removed from a 16-character App Password.
- Hardened SMTP transport with the operating system trust store, explicit EHLO → STARTTLS → EHLO negotiation, authentication, and NOOP readiness verification.
- Added smtp-check (no message) and smtp-test (one clearly labelled self-message) operational commands. Neither prints the credential.
- Passed SMTP variables through Docker Compose while keeping the ignored local environment and generated image layers separate.
- Added PyMuPDF to backend development dependencies so offer-letter rendering tests install and run reproducibly.
- Added reproducible ESLint configurations and dependencies for both web applications; fixed the single registration lint defect.
- Added root lint/test helper scripts, refreshed documentation, and normalized source line endings so git diff --check is clean.

Verification matrix

Check	Status	Evidence
Project layout	PASS	backend/, frontend/console/, and frontend/registration/ are present; referenced Firebase, workspace, and Compose paths resolve.
Python lock/install	PASS	uv lock check passed; 77 packages resolved; byte compilation completed.
Backend lint	PASS	Ruff completed with no findings.
Backend local suite	PASS	97 passed; 0 failed; 213 emulator-dependent tests skipped; one upstream deprecation warning.
API process	PASS	Uvicorn application startup completed and /health returned status=ok, version=1.0.0, env=development.
OpenAPI	PASS	41 paths / 54 operations; 0 missing and 0 duplicate operation IDs; /docs returned HTTP 200.
SMTP schema	PASS	Local and production-template settings validate; Gmail endpoint/security/sender alignment and 16-character secret shape pass.
SMTP local delivery	PASS	In-process SMTP catcher received the message, multipart body, inline brand image, and PDF attachment.
SMTP TLS paths	PASS	STARTTLS and implicit SSL use a verified default TLS context; readiness authenticates and issues NOOP.
Credential hygiene	PASS	Secret is in the ignored local environment only; scan found 0 occurrences elsewhere in source; output/report are redacted.
Console tests	PASS	Vitest: 1 file / 20 tests passed.

Check	Status	Evidence
Frontend lint	PASS	Console and registration ESLint runs completed with no findings.
Production builds	PASS	Console and registration Vite builds completed; each compiled API origin matches the configured Cloud Run URL.
Build serving	PASS	Both preview index pages and their JavaScript assets returned HTTP 200.
Config syntax	PASS	Compose/workspace YAML and Firebase/index JSON parse; all referenced source/rules paths exist.
Live Gmail auth	BLOCKED	smtp.gmail.com could not be resolved from this workspace (socket.gaierror), before TLS/login.
Live self-delivery	BLOCKED	smtp-test stopped at readiness; no message was sent and inbox receipt was not tested.
Firebase E2E	NOT RUN	213 Firestore/Storage tests require emulator binaries; downloads are blocked and no cached binaries are available.
Cloud Run SMTP	NOT RUN	Source template and commands are ready, but the deployed service was not mutated or inspected.

How to read these results

NO LOCAL FAILURES

Every check that could execute inside this workspace passed. The four non-pass rows are external verification gates, not observed application defects.

The most important distinction is that a network-level block does not validate a credential. Live Gmail authentication and inbox delivery remain pending until the readiness and self-test commands complete outside this restricted environment.

API and application evidence

API surface

The generated OpenAPI document contains 54 operations across 41 paths. Method counts: 26 GET, 20 POST, 4 DELETE, 3 PATCH, and 1 PUT. Tagged domains include Auth, Admin, Applications, Students, Batches, Attendance, Payments, Reports, Notifications, Announcements, Settings, and Public registration.

Runtime smoke result

```
GET /health → 200 { status: ok, version: 1.0.0, env: development }
```

This verifies application import, configuration parsing, middleware construction, router registration, local socket serving, health serialization, documentation serving, and OpenAPI generation. It does not substitute for data-backed endpoint tests.

Frontend evidence

Application	Lint	Tests	Build	Preview
Staff console	Pass	20/20	Pass	200 / 200
Registration	Pass	No suite	Pass	200 / 200

Advisory: both production builds report JavaScript chunks above Vite's 500 kB advisory threshold (console 1,058.35 kB; registration 638.15 kB before gzip). This is a performance optimization opportunity, not a build or correctness failure.

SMTP configuration and safety

Setting	Configured value
Server	smtp.gmail.com
Transport	STARTTLS on port 587
Username / envelope sender	dveininnovation@gmail.com
From / Reply-To	Dvein Innovations <dveininnovation@gmail.com>
Credential	Present locally; 16-character App Password shape; value redacted
Production credential	Not stored in .env.production; mount SMTP_PASSWORD from Secret Manager

External completion checklist

ACTION 1

Run the no-send SMTP readiness command from a machine or deployment that can reach smtp.gmail.com:587.

```
cd backend
uv run python -m app.cli smtp-check
```

Expected: SMTP : OK – TLS connection and authentication succeeded. An authentication error means the App Password was invalid/revoked or 2-Step Verification is unavailable; a connection/DNS error remains a network problem.

ACTION 2

Only after Action 1 passes, send one labelled message to the configured sender and confirm it arrives in the inbox (or Spam).

```
uv run python -m app.cli smtp-test
```

ACTION 3

Apply the non-secret SMTP values and Secret Manager reference to the Cloud Run service, then repeat smtp-check inside that runtime.

Use the first-time/rotation commands in README.md under “Configuring or rotating Gmail SMTP.” The App Password must be a secret version and must never be baked into an image or committed file.

ACTION 4

Start Firestore and Storage emulators on a Java 21+ machine, then run the full backend suite and require all 310 tests to execute with zero skips/failures.

```
pnpm emulators:fresh
cd backend && uv run pytest
```

Known advisories

- FastAPI/Starlette emits one deprecation warning from the installed TestClient integration. It is upstream of application code and did not affect results.
- The two browser bundles exceed Vite's default chunk-size advisory. Introduce route-level dynamic imports when performance work is scheduled.
- Because an App Password was shared in chat, rotate it before production if strict credential handling is required, then update the ignored local environment and Secret Manager together.

Official references

Google Workspace Help — [Send email from a printer, scanner, or app](#). Google documents smtp.gmail.com, SSL port 465, TLS port 587, the complete email address, and an App Password for authenticated SMTP.

Google Account Help — [Sign in with app passwords](#). Google documents App Passwords as 16-character passcodes that require 2-Step Verification and may be revoked after an account-password change.

Final technical position

The source package is ready. Do not mark Gmail delivery, Cloud Run SMTP, or Firebase-backed API integration as production-verified until Actions 1–4 pass in an unrestricted environment.